

$$\int_1^{1.5} f(x) dx = \frac{1}{4} \quad \int_{1.5}^3 f(x) dx = -\frac{1}{2} \quad \int_3^4 f(x) dx = 1$$

$$\int_1^3 f(x) dx = \int_1^{1.5} f(x) dx + \int_{1.5}^3 f(x) dx = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

$$\int_1^4 f(x) dx = \underbrace{\int_1^{1.5} f(x) dx}_{\frac{1}{4}} + \underbrace{\int_{1.5}^3 f(x) dx}_{-\frac{1}{2}} + \underbrace{\int_3^4 f(x) dx}_1$$

$$\int x \cos(x^2) dx$$

$$u = x^2$$

$$du = 2x \underline{\underline{dx}}$$

$$\underline{\underline{du = 2x}}$$

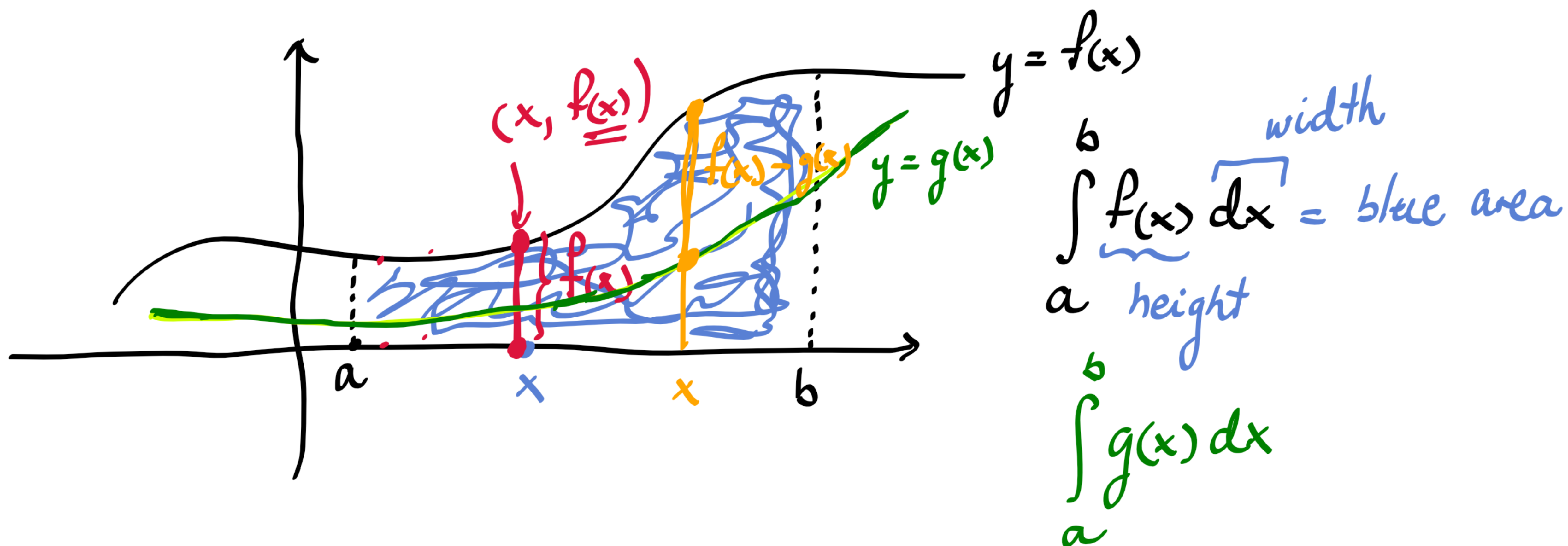
$$\int_0^{2\pi} \underbrace{\sin x}_u \underbrace{\cos x dx}_{du} = \int_{\cancel{0} \sin(0)}^{\cancel{2\pi} \sin(2\pi)} u du = \underbrace{\frac{1}{2} u^2}_\frac{1}{2} \Big|_0^{2\pi} = \underline{\underline{\frac{1}{2} (\sin x)^2 \Big|_0^{2\pi} = 0}}$$

Not equal!

$u = \sin x$
 $du = \cos x dx$

$$\left\{ \begin{aligned} \int \sin x \cos x dx &= \int u du = \frac{1}{2} u^2 + C = \frac{1}{2} (\sin x)^2 + C \\ \int_0^{2\pi} \sin x \cos x dx &= \frac{1}{2} (\sin x)^2 \Big|_0^{2\pi} = 0 \end{aligned} \right.$$

Section 2.1 - Areas between curves

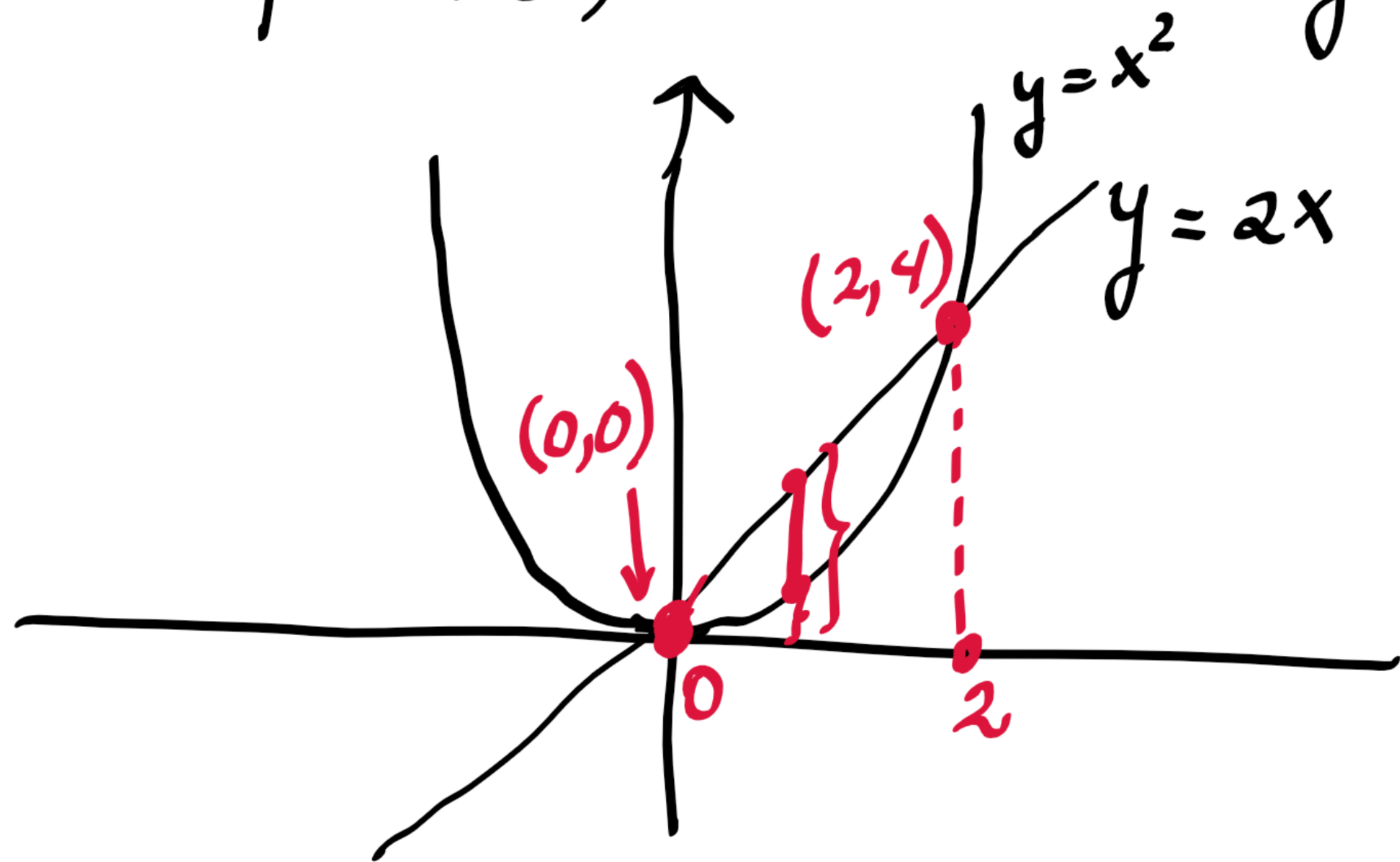


Area between the graphs of $f(x)$ and $g(x)$:

$$\int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b (f(x) - g(x)) dx$$

for all $x \in [a, b]$,
 $f(x) \geq g(x)$

Example : Find area of region enclosed by the graphs of $f(x) = x^2$ and $g(x) = 2x$.



$$\begin{aligned} \left. \begin{aligned} y &= x^2 \\ y &= 2x \end{aligned} \right\} & \begin{aligned} x^2 &= 2x \\ x^2 - 2x &= 0 \\ x(x-2) &= 0 \end{aligned} \\ & \begin{aligned} x &= 0 \quad \text{or} \quad x-2=0 \\ & \underline{x=2} \end{aligned} \end{aligned}$$

$$\left. \begin{aligned} g(1) &= 2 \\ f(1) &= 1 \end{aligned} \right\} g(x) \geq f(x)$$

$$\int_0^2 (2x - x^2) dx$$

$$\int_0^2 (x^2 - 2x) dx$$

2) Find the area of the region bounded by the graphs of $f(x) = x^2 - 2x - 3$ and $g(x) = x - 3$ between -1 and 1.

$$\begin{cases} y = x^2 - 2x - 3 \\ y = x - 3 \end{cases}$$

$$x - 3 = x^2 - 2x - 3$$

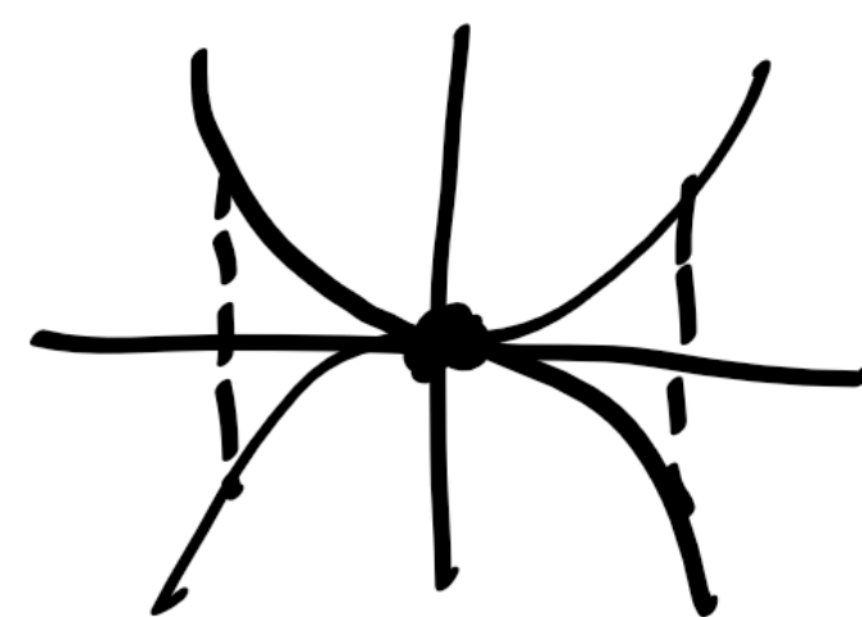
$$x^2 - 3x = 0$$

$$x(x - 3) = 0$$

$$\underline{x = 0} \quad \text{or} \quad x - 3 = 0$$

$$\downarrow \quad \underline{x = 3}$$

$$(0, -3) \quad (3, 0)$$



$$\begin{array}{ccccccc} -1 & & 0 & & 1 & & 3 \\ & \swarrow & & \searrow & & & \\ & -\frac{1}{2} & & \frac{1}{2} & & & \end{array}$$

$$\int_{-1}^0 (x^2 - 2x - 3 - (x - 3)) dx + \int_0^1 ((x - 3) - (x^2 - 2x - 3)) dx$$

$$\int_{-1}^0 (x^2 - 2x - 3 - x + 3) dx + \int_0^1 (x - 3 - x^2 + 2x + 3) dx$$

$$\int_{-1}^0 (x^2 - 3x) dx + \int_0^1 (3x - x^2) dx$$

Evaluating the definite integral, we get the area.

$$f(x) = x^2 - 2x - 3$$

$$\begin{cases} y = a(x-h)^2 + k \\ x = a(y-k)^2 + h \end{cases}$$

$$y = x^2 - 2x - 3 + 1 - 1$$

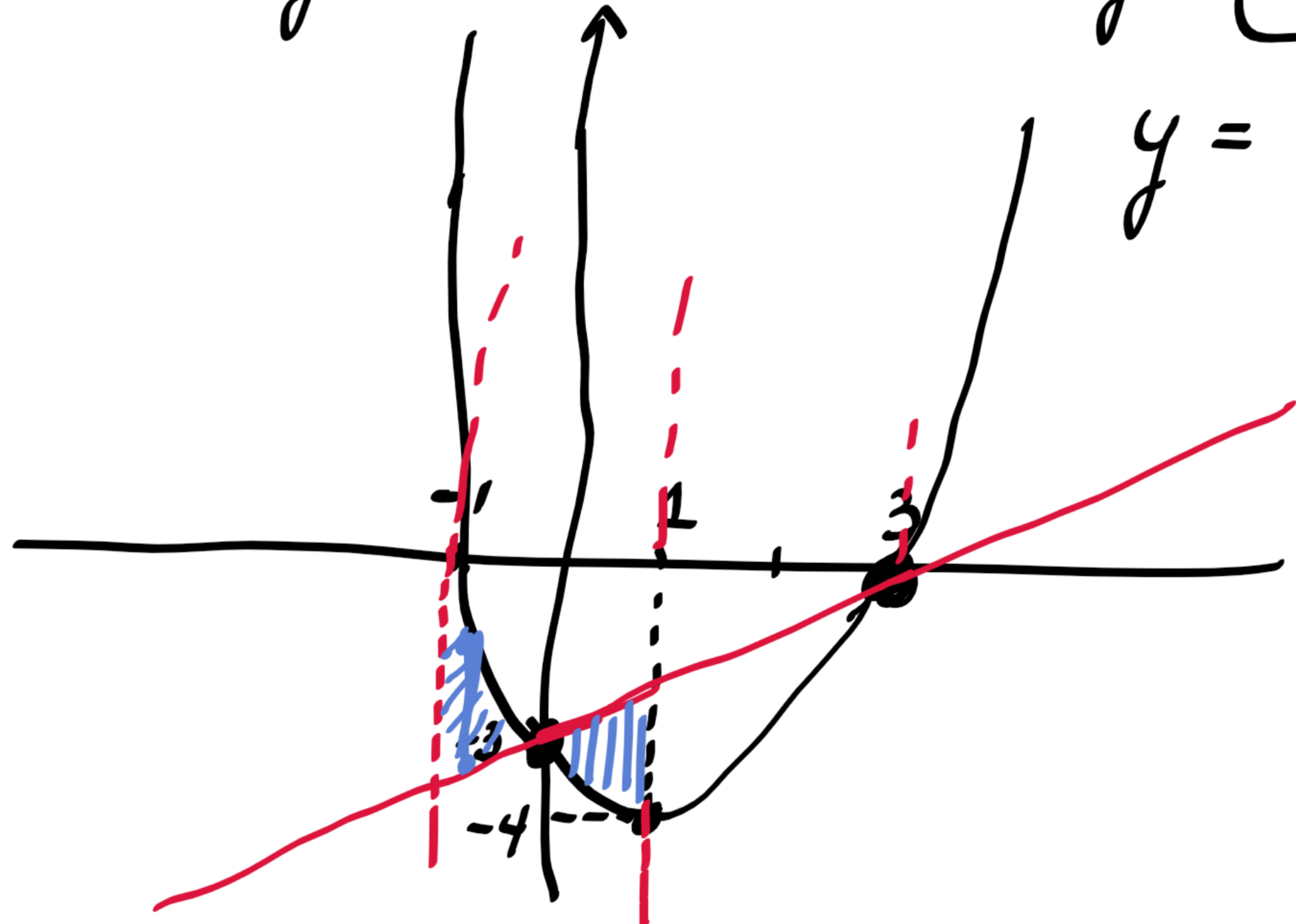
$$\frac{-2}{2} = -1 \quad (-1)^2 = 1$$

$$y = x^2 - 2x + 1 - 1 - 3$$

$$y = (x-1)^2 - 4 \quad (1, -4)$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$



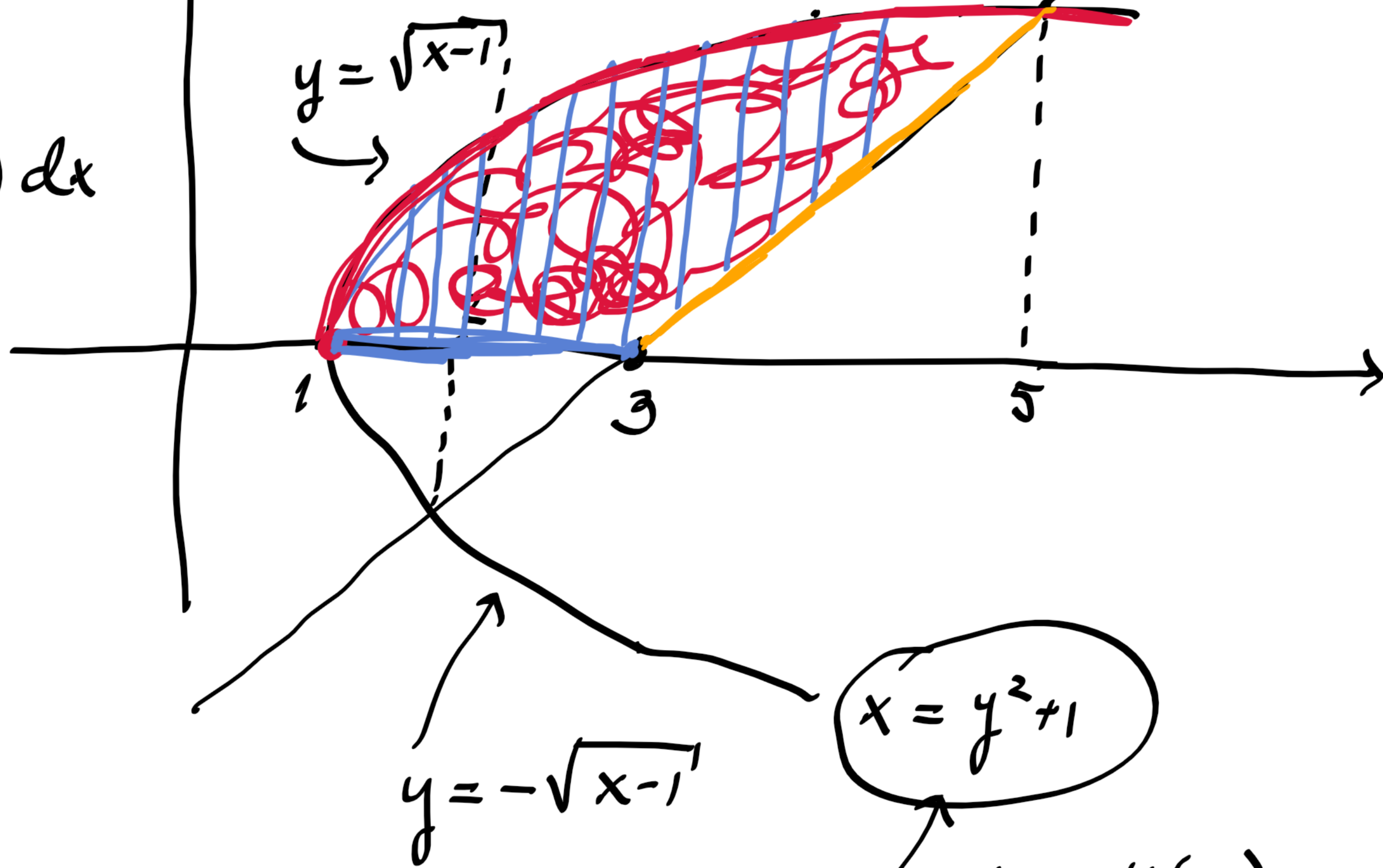
functions of x

$$\int_1^3 (\text{red} - \text{blue}) dx$$

$$+ \int_3^5 (\text{red} - \text{orange}) dx$$

$$\int_1^3 (\sqrt{x-1} - 0) dx + \int_3^5 (\sqrt{x-1} - (x-3)) dx$$

$y = x-3$



$$x = y^2 + 1$$

$$x - 1 = y^2$$

$$\sqrt{x-1} = |y|$$

$$y = \sqrt{x-1} \quad \text{or} \quad y = -\sqrt{x-1}$$

$$x = y^2 + 1$$

$$x = u(y)$$